# Čencov's Theorem in Resolution Geometry

## Full Derivation from Descent, Quantitative Stability, and Finite-Scale Recovery

Matěj Rada

Companion to *Resolution Geometry* (print edition)

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**Abstract**

Resolution Geometry (RG) states that what observations determine is a quotient, and that every admissible quantity must *descend* through that quotient. We show that this descent principle *is* Čencov's hypothesis, and we derive Čencov's theorem from it in full.

The key lemma is that a Markov map admits a Markov left inverse, i.e. it is a lossless re-description, iff its columns have disjoint supports, i.e. iff it is a congruent splitting. A continuous positive metric that descends through lossless re-descriptions is therefore $c$ times Fisher, with a single constant $c$ across all dimensions. Conversely, Fisher descends and contracts under every channel, so for continuous metrics "descent" and "monotonicity" coincide. Each hypothesis is shown to be necessary by an explicit counterexample, except continuity, which remains an open question.

In quantum theory the same descent holds for every Petz metric but selects none of them (executed: 100/100 invariance, relative spread up to 16.6 on noncommuting tangents). That is exactly why RG must declare a task there.

RG therefore exists because of Čencov in a precise sense:

- its classical metric layer is *forced* by descent;
- its remaining machinery is needed where Čencov provably stops, namely global experiment equivalence, finite divergences, singular strata and quantum states.

The uniqueness of Fisher–Rao geometry does not make a local information tensor a complete invariant of a statistical experiment. We develop a scoped theorem package around this distinction. First, a continuous, permutation-invariant family of positive quadratic metrics on all finite open simplices, whose distortion under every congruent splitting is at most $e^{\pm\varepsilon}$, is within $e^{\pm 2\varepsilon}$ of one normalized Fisher family. A common uniform refinement proves this bound. Approximate permutation symmetry with distortion $e^{\pm\delta}$ gives $e^{\pm 2(\varepsilon+\delta)}$; the splitting premise alone gives $e^{\pm 4\varepsilon}$. The exact characterization is recovered at $\varepsilon = 0$. Second, the Fisher contraction defect is a conditional score variance. Its finite counterpart controls a reference-Bayes recovery map; its integral along an affine mixture is exactly the relative-entropy loss. A pathwise posterior-variation bound supplies a common recovery map and an explicit deficiency certificate on connected positive families. Reference-consistent recoveries compose exactly, while information losses telescope. We derive stability statements under square-root derivative control and spectral margins, and a finite-partition extension on dominated tangent directions. Counterexamples separate Fisher-null tangents from exact observational fibres, local metrics from global experiment equivalence, small deficiency from derivative control, and spectral selection from unrestricted approximate admissibility. A limited quantum section proves the corresponding entropy-Hessian integral for the BKM metric without claiming a unique quantum metric or a classical-style quantum recovery bound. All positive claims have proofs or explicitly identified imported inputs. Reproducible synthetic, exact-rational, and matrix checks supplement but do not replace the proofs. No priority claim or exhaustive literature clearance is made.

**Status.** Print edition. The exact derivation (Section 3) and every quantitative statement are proved under their displayed assumptions. The paper is not independently refereed or formally verified. The exact characterization is Čencov's; the RG glue (Lemma 3.2, Theorem 3.4) and the quantitative package are the proposed contribution, and their priority remains unreviewed.

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# 1 Question, contribution, and claim discipline

Čencov's theorem is a characterization of a *family of metrics across statistical spaces*, not merely a statement at one parameter value. Its output is nevertheless a local quadratic object. Thus "Fisher is locally limited" should not be read as a defect in the characterization. It means that an infinitesimal tensor does not retain every finite distinction, likelihood embedding, support change, or protocol choice.

Resolution Geometry (RG) separates equality of observation laws, comparison by randomization, local metric selection, and declared resolution gates [8, 9]. We use that separation to ask four precise questions:

1. Does approximate sufficient-embedding invariance force an approximately Fisher metric?
2. What does the amount of Fisher loss certify about finite recovery?
3. Which local-to-global and multiscale implications are valid, and under which hypotheses?
4. Which desired closures are false and must instead become explicit no-go results?

The answer is a quantitative extension of the *characterization setting*, together with a surrounding recovery calculus. It is not a replacement metric satisfying the unchanged exact classical axioms.

**Provenance and attribution.** The exact characterization, data processing, conditional-expectation interpretation of scores, classical sufficiency, $f$-divergences, comparison of experiments, and quantum monotone-metric classification are established mathematical territory [1, 2, 3, 4, 5, 7]. We rederive the finite results needed here to make the logical chain inspectable. A proof given here is not a claim that the result was first discovered here. The proposed contribution is the explicit quantitative package and its RG-facing assumption ledger; even that contribution needs a dedicated priority review.

**Claim classes.** (C1) Established imports are labeled as such. (C2) Theorems, propositions, and corollaries are analytic consequences proved below. (C3) Computations are finite implementation checks on synthetic mathematical examples. (C4) No-go and open-problem statements prevent promotion outside the proved scope. There are no "empirical universal theorems."

# 2 Finite experiments and elementary inequalities

Let $\Delta_n^\circ = \{p \in \mathbb{R}^n : p_i > 0, \sum_i p_i = 1\}$ and $T_n = \{u : \sum_i u_i = 0\}$. Write

$$F_p(u,v) = \sum_i \frac{u_i v_i}{p_i}, \qquad F_p(u) = F_p(u,u). \tag{1}$$

A channel $K : X \to Y$ is a column-stochastic matrix $K_{yx} \geq 0$, $\sum_y K_{yx} = 1$, independent of the parameter. Delete rows identically zero. Then $Kp > 0$ whenever $p > 0$. A one-point output is permitted and has zero tangent metric; set $h^{(1)} = 0$ when needed. All logarithms are natural and

$$\mathrm{TV}(p,q) = \tfrac{1}{2}\sum_x |p_x - q_x|, \quad D(p\|q) = \sum_x p_x \log(p_x/q_x), \quad \chi^2(p\|q) = \sum_x \frac{(p_x - q_x)^2}{q_x}.$$

**Definition 2.1** (Congruent splitting)**.** Partition an $m$-point output alphabet into nonempty blocks $B_1, \ldots, B_n$. Choose positive weights $a_{y|i}$ summing to one in each block. Define $E_{yi} = a_{y|i}$ for $y \in B_i$ and zero otherwise. The deterministic block-merging map $M$ satisfies $ME = I_n$. Such $E$ is a congruent splitting.

**Lemma 2.2** (Log-sum, processing, and Pinsker)**.** *For positive laws, $D(Kp\|Kq) \le D(p\|q)$ and*

$$2\,\mathrm{TV}(p,q)^2 \le D(p\|q). \tag{2}$$

*More generally $D_f(p\|q) = \sum_x q_x f(p_x/q_x)$ contracts for every convex $f$ with $f(1) = 0$.*

*Proof.* For each $y$, put $\alpha_{x|y} = q_x K_{yx}/(Kq)_y$. These weights sum to one and $(Kp)_y/(Kq)_y = \sum_x \alpha_{x|y} p_x/q_x$. Jensen's inequality, followed by summation in $y$, proves the $f$ statement. Taking $f(t) = t\log t$ proves relative-entropy processing and the log-sum inequality. For Pinsker let $A = \{x : p_x \ge q_x\}$, $a = p(A)$, and $b = q(A)$. Then $a - b = \mathrm{TV}(p,q)$. Merging $A$ and its complement gives $D(p\|q) \ge d(a\|b)$, the binary relative entropy. For fixed $b$, this function has value and first derivative zero at $a = b$, and second derivative $1/[a(1-a)] \ge 4$. Taylor's integral formula gives $d(a\|b) \ge 2(a-b)^2$. Endpoint cases follow by limits. □

## 3 RG descent is Čencov's hypothesis: the exact derivation

RG's basic rule is that a quantity attached to a protocol is admissible only if it *descends* through observational equivalence. Two records that determine each other by known randomizations carry the same information, so they must receive the same metric. This section turns that rule into Čencov's hypothesis and derives the theorem from it.

**Definition 3.1** (Lossless re-description)**.** A Markov map $K : \Delta_n \to \Delta_m$ (a column-stochastic matrix) is a *lossless re-description* if some Markov map $M : \Delta_m \to \Delta_n$ satisfies $MK = I_n$. Two experiments related by a lossless re-description are Blackwell equivalent: each is a garbling of the other.

**Lemma 3.2** (Lossless re-descriptions are congruent splittings)**.** *$K$ is a lossless re-description iff its columns have pairwise disjoint supports. In that case, after deleting outputs that never occur, $K$ is a congruent splitting composed with a relabeling of outputs, and $M$ may be taken to be the merge.*

*Proof.* Suppose $MK = I$. For each input $i$, $1 = (MK)_{ii} = \sum_y M_{iy} K_{yi}$, with $\sum_y K_{yi} = 1$ and $0 \le M_{iy} \le 1$. A convex combination of numbers at most one equals one only if every number with positive weight equals one. Hence $K_{yi} > 0$ implies $M_{iy} = 1$. Since the columns of $M$ sum to one, $M_{jy} = 0$ for $j \ne i$. If also $K_{yj} > 0$ for some $j \ne i$, the same argument would give $M_{jy} = 1$, a contradiction. So the supports are disjoint.

Conversely, with disjoint supports $B_i$, let $M$ send every $y \in B_i$ to $i$ (unused outputs anywhere). Then $MK = I$. Writing $a_{y|i} = K_{yi}$ on $B_i$ exhibits $K$ as a congruent splitting with blocks $B_i$, up to the order of outputs. Singleton blocks give permutations. □

**Definition 3.3** (RG information metric)**.** An *RG information metric* assigns to every $n \ge 2$ and every $p \in \Delta_n^\circ$ a quadratic form $h_p^{(n)}$ on $T_n$ such that:

(R1) *(descent)* $h_{Kp}^{(m)}(Ku) = h_p^{(n)}(u)$ for every lossless re-description $K$, with unused outputs deleted;

(R2) *(continuity)* $p \mapsto h_p^{(n)}(u)$ is continuous for each $u$;

(R3) *(positivity)* $h_p^{(n)}(u) > 0$ for $u \ne 0$.

**Theorem 3.4** (Čencov's theorem from RG descent)**.** *Every RG information metric equals $cF$ with the single constant $c = h_{U_2}^{(2)}((1,-1))/4 > 0$, in every dimension, at every interior law, as a bilinear form. Conversely $cF$ is an RG information metric for every $c > 0$, and it contracts under every Markov map. Hence, for continuous positive metric families,*

*descent through lossless re-descriptions $\iff$ monotonicity under all Markov maps $\iff$ $h = cF$.*

*Proof. Step 1 (isotropy at the uniform law).* Permutations are lossless re-descriptions (Lemma 3.2, singleton blocks). By (R1) the form at $U_m$ commutes with all permutations, so Lemma 4.2 (whose proof uses only permutation invariance) gives $h_{U_m} = c_m F_{U_m}$, with $c_m > 0$ by (R3).

*Step 2 (one constant for all dimensions).* Splitting each outcome of $U_m$ into two halves is lossless and sends $U_m$ to $U_{2m}$. Fisher is splitting-invariant (Step 6), so (R1) gives $c_{2m} = c_m$. Splitting each outcome of $U_2$ into $m$ equal parts sends $U_2$ to $U_{2m}$, so $c_{2m} = c_2 = c$. Hence $c_m = c$ for every $m$.

*Step 3 (rational laws).* Let $p_i = k_i/N$ with positive integers $k_i$. Splitting outcome $i$ into $k_i$ equal parts is lossless, sends $p$ to $U_N$, and sends $u$ to a vector $Eu$ with $F_{U_N}(Eu) = F_p(u)$. By (R1) and Steps 1–2, $h_p(u) = h_{U_N}(Eu) = cF_{U_N}(Eu) = cF_p(u)$.

*Step 4 (all laws).* Positive rational laws are dense in $\Delta_n^\circ$. By (R2), and the continuity of $F$, $h_p(u) = cF_p(u)$ for every interior $p$.

*Step 5 (bilinear form).* A symmetric bilinear form is determined by its quadratic form, by polarization.

*Step 6 (converse).* Under a splitting, block $i$ contributes $\sum_{y\in B_i}(a_{y|i}u_i)^2/(a_{y|i}p_i) = u_i^2/p_i$, so $F$ is exactly invariant. Every Markov map contracts $F$ because the contraction defect is a conditional score variance (Theorem 5.1). A metric that is monotone under all Markov maps satisfies (R1), because $K$ and its left inverse $M$ are both Markov: $h_p(u) \geq h_{Kp}(Ku) \geq h_{MKp}(MKu) = h_p(u)$. □

*Remark* 3.5 (Two descents, and the print's L29). The print's No-Go (L29) says that descent *within a fixed family* does not select a metric. On the Bernoulli family with outcome exchange, every $a(p)\,dp^2$ with $a(p) = a(1-p)$ is law-natural. That remains true. Condition (R1) is a different and stronger descent: it holds *across experiments*, through every lossless re-description between simplices of different dimension. Splitting a Bernoulli outcome into two forces $a(p) = c/[p(1-p)]$. Theorem 3.4 therefore upgrades the print's imported Čencov selection: H-mono may be replaced by experiment-level descent together with continuity, and the selection is derived rather than imported.

**Corollary 3.6** (The classical metric layer of RG is forced)**.** *Any RG construction that assigns a continuous positive local information geometry to protocol states, and respects descent, uses Fisher up to one global constant. This is the content of the print's Fisher rungs (L28–L29) and of the unit-anchor rule: $c$ is the only freedom left, and it is a choice of units.*

**Proposition 3.7** (Each hypothesis is necessary, except possibly continuity)**.**

1. Permutations alone are not enough. *$h_p = (\sum_i p_i^2)F_p$ is permutation-invariant. Under random congruent splittings it violates descent by up to* log-*ratio* 1.05 *(executed, G2).*

2. Descent must connect dimensions. *$h^{(n)} = (1{+}0.1n)F$ satisfies descent within each dimension but fails across a splitting $2 \to m$ (G2).*

3. Positivity. *The zero form satisfies (R1) and (R2) and is monotone, but it is degenerate.*

4. Continuity: open. *Splitting and merging relations connect every binary law to $U_2$ (for example, $(\beta, 1-\beta)$ and $U_2$ are both merges of $(\beta, \frac{1}{2}-\beta, \frac{1}{2})$). However, at an irrational law the induced relations constrain the metric only on finitely many tangent lines of intermediate laws, whose remaining values are themselves unknown. Whether a discontinuous non-Fisher family satisfying (R1) and (R3) exists is not settled here (Open Problem 17.5).*

## 4 Quantitative metric selection

**Assumption 4.1** (Uniform approximate sufficient invariance)**.** For every $n \geq 2$, $h^{(n)}$ is a continuous positive-definite quadratic metric on $\Delta_n^\circ$, exactly invariant under permutations. There is

one $\varepsilon \geq 0$ such that for *every* congruent splitting $E : n \to m$, every $p \in \Delta_n^\circ$, and every $u \in T_n$,

$$e^{-\varepsilon} h_p^{(n)}(u) \leq h_{Ep}^{(m)}(Eu) \leq e^{\varepsilon} h_p^{(n)}(u). \tag{3}$$

The same $\varepsilon$ applies in all dimensions and at all interior laws.

**Lemma 4.2** (Uniform-law isotropy). *At $U_m = (1/m, \ldots, 1/m)$ there is $c_m > 0$ with $h_{U_m}^{(m)} = c_m F_{U_m}$.*

*Proof.* Extend the bilinear form to $\mathbb{R}^m$ by composing with the Euclidean projection onto $T_m$. Its representing symmetric matrix commutes with every permutation. Permutations force all diagonal entries to be equal and all off-diagonal entries to be equal. The matrix therefore has the form $aI + b\mathbf{1}\mathbf{1}^\top$. On $T_m$ it is $aI$, with $a > 0$. Since $F_{U_m}(u) = m \sum_i u_i^2$, set $c_m = a/m$. □

**Theorem 4.3** (Quantitative Čencov stability under exact permutation symmetry). *Under Assumption 4.1, define*

$$c = \frac{h_{U_2}^{(2)}((1,-1))}{4} > 0.$$

*For every $n \geq 2$, every $p \in \Delta_n^\circ$, and every $u \in T_n$,*

$$e^{-2\varepsilon} c F_p(u) \leq h_p^{(n)}(u) \leq e^{2\varepsilon} c F_p(u). \tag{4}$$

*The constant 2 in the exponent is a proved bound, not an asserted optimal constant.*

*Proof.* First let $p_i = k_i/N$ with positive integers $k_i$ and $\sum_i k_i = N$. Split outcome $i$ into $2k_i$ equally weighted outcomes. The resulting $E$ sends $p$ to $U_{2N}$ and sends $u_i$ to $2k_i$ copies of $u_i/(2k_i)$. Hence

$$F_{U_{2N}}(Eu) = 2N \sum_i 2k_i \left(\frac{u_i}{2k_i}\right)^2 = \sum_i \frac{u_i^2}{p_i} = F_p(u).$$

By Lemma 4.2, $h_{U_{2N}}(Eu) = c_{2N} F_p(u)$. Independently split each outcome of $U_2$ into $N$ equal outcomes. Applying (3) to $(1,-1)$ gives $e^{-\varepsilon} c \leq c_{2N} \leq e^{\varepsilon} c$. Applying (3) to the first splitting and solving for $h_p(u)$ gives $e^{-\varepsilon} c_{2N} F_p(u) \leq h_p(u) \leq e^{\varepsilon} c_{2N} F_p(u)$. Combining the two inequalities proves (4) at rational $p$. Positive rational laws are dense in each open simplex. Continuity of $h$ and $F$ extends the result to every $p$, with $u$ fixed. The zero tangent is immediate. □

**Theorem 4.4** (Approximate symmetry and a symmetry-free bound). *Retain the continuity, positivity, and uniform splitting assumption, but replace exact permutation invariance by*

$$e^{-\delta} h_{U_m}(u) \leq h_{U_m}(\Pi u) \leq e^{\delta} h_{U_m}(u)$$

*for every uniform law and every permutation $\Pi$, with one dimension-uniform $\delta \geq 0$. With the same binary normalization $c$,*

$$e^{-2(\varepsilon+\delta)} c F_p \preceq h_p \preceq e^{2(\varepsilon+\delta)} c F_p.$$

*If no permutation hypothesis is imposed separately, the splitting assumption itself gives $\delta = \varepsilon$, hence a universal $e^{\pm 4\varepsilon}$ bound.*

*Proof.* At $U_m$ average the quadratic form over all permutations. The averaged form is isotropic by Lemma 4.2, say $\bar{h}_{U_m} = \bar{c}_m F_{U_m}$. The assumed inequalities imply $e^{-\delta} \bar{h}_{U_m} \preceq h_{U_m} \preceq e^{\delta} \bar{h}_{U_m}$. On the binary tangent, every quadratic form is automatically permutation invariant, so $\bar{c}_2 = c$. In the two-refinement proof of Theorem 4.3, comparison with the binary splitting now gives $e^{-(\varepsilon+\delta)} c \leq \bar{c}_{2N} \leq e^{\varepsilon+\delta} c$. The rational-law splitting gives $e^{-(\varepsilon+\delta)} \bar{c}_{2N} F_p \preceq h_p \preceq e^{\varepsilon+\delta} \bar{c}_{2N} F_p$. Combining and passing to irrational laws proves the claim. A permutation is a congruent splitting with singleton blocks, so the uniform splitting premise includes its $e^{\pm\varepsilon}$ distortion and supplies $\delta = \varepsilon$ without a separate symmetry axiom. □

**Corollary 4.5** (Exact characterization and no exact replacement). *At $\varepsilon = 0$, the only such metric families are $h = cF$. Conversely $cF$ satisfies exact splitting invariance and contracts under all channels. In particular an exact alternative to Fisher must change at least one hypothesis.*

*Proof.* The first statement is Theorem 4.3. Exact splitting invariance follows by summing $a_{y|i}^2 u_i^2/(a_{y|i}p_i)$ over each block. General contraction follows from Theorem 5.1 below. A metric family monotone under all channels is invariant under congruent splittings because both $E$ and its left inverse $M$ are channels. Thus the familiar monotone formulation lies inside this characterization. □

**Corollary 4.6** (Pullback, nulls, and distance stability). *For a $C^1$ law map $p : \Theta \to \Delta_n^\circ$, the pullback satisfies (4) with $u = Dp_\theta v$, and $\mathrm{rad}(p^*h)_\theta = \ker Dp_\theta = \mathrm{rad}(p^*F)_\theta$. On a common connected regular quotient, intrinsic distances obey*

$$e^{-\varepsilon}\sqrt{c}\, d_F \le d_h \le e^{\varepsilon}\sqrt{c}\, d_F.$$

*Proof.* Substitution gives the quadratic bounds. A positive quadratic form vanishes exactly at zero, proving the radical identity. Taking square roots, integrating along each admissible path, and taking infima proves the distance bounds. The quotient and path class must be the same on both sides. □

**Proposition 4.7** (Selection uncertainty is not loss uncertainty). *Let $a = e^{2\varepsilon} - 1$. For $\Delta_{p,K}^h(u) = h_p(u) - h_{Kp}(Ku)$,*

$$\left|\Delta_{p,K}^h(u) - c\mathcal{D}_{p,K}(u)\right| \le ca\{F_p(u) + F_{Kp}(Ku)\} \le 2caF_p(u),$$

*where $\mathcal{D}_{p,K}(u) = F_p(u) - F_{Kp}(Ku)$. In particular approximate splitting invariance alone does not assert that $\Delta^h$ is positive.*

*Proof.* Equation (4) implies $|h - cF| \le caF$ on each quadratic evaluation. Apply the triangle inequality to the two evaluations and Fisher contraction to the second. The assumptions contain no monotonicity inequality for arbitrary non-sufficient channels, so none is inferred for $h$. □

**Example 4.8** (Nonempty approximate class). The family $h_p^{(n)} = \exp(\varepsilon \sum_i p_i^2) F_p$ is permutation invariant and satisfies (3): both $\sum_i p_i^2$ and $\sum_y (Ep)_y^2$ lie in $(0, 1]$, so their difference has absolute value at most one. For $\varepsilon > 0$ the multiplier is not constant. Exact uniqueness has therefore not been silently assumed in the quantitative theorem.

*No-Go* 4.9 (A finite invariance panel cannot certify the universal premise). Without a declared regularity modulus, no finite collection of metric evaluations certifies (3) everywhere.

*Proof.* In the binary simplex choose an open interval disjoint from every queried binary law, its exchanged law, and $U_2$. Put a nonnegative smooth bump there and its reflected interval, obtaining a symmetric function $b$ vanishing at every queried point. Set $h^{(2)} = (1 + Mb)F$ and set $h^{(n)} = F$ for $n \ne 2$. Every queried metric value is unchanged. At an unqueried point where $b > 0$, a splitting into four outcomes has an arbitrarily large invariance violation as $M \to \infty$. The same construction avoids sufficiently small neighborhoods and every finite jet at the queried points. □

## 5 Exact information-loss calculus

For $p > 0$, define the score $s_x = u_x/p_x$. Under the joint law $p_x K_{yx}$ the output score is

$$\bar{s}_y = \frac{(Ku)_y}{(Kp)_y} = \mathbb{E}_p[s(X) \mid Y = y]. \tag{5}$$

**Theorem 5.1** (Fisher defect is conditional score variance). *For every finite positive experiment and parameter-independent channel,*

$$\mathcal{D}_{p,K}(u) = \sum_{x,y} p_x K_{yx} (s_x - \bar{s}_y)^2 \geq 0. \tag{6}$$

*Equality holds iff $s_x = \bar{s}_y$ on every edge $K_{yx} > 0$.*

*Proof.* Expanding the square gives $\sum_x p_x s_x^2 - 2\sum_y (Kp)_y \bar{s}_y^2 + \sum_y (Kp)_y \bar{s}_y^2$, which is precisely $F_p(u) - F_{Kp}(Ku)$. All edge weights in the sum are positive on the support of $K$, so equality is equivalent to the displayed edge condition. □

**Corollary 5.2** (Exact lossless sectors). *At fixed $p$, the zero-defect tangents form a linear subspace. For finitely many declared channels, their common zero-defect sector is the intersection of these subspaces and is their unique greatest common lossless subspace.*

*Proof.* The edge equations in Theorem 5.1 are homogeneous linear equations in $u$. Their simultaneous solution set is a kernel. Every common lossless subspace is contained in the intersection, and the intersection itself is lossless. □

**Proposition 5.3** (Defect cocycle and independent products). *For composable channels $K, L$,*

$$\mathcal{D}_{p,LK}(u) = \mathcal{D}_{p,K}(u) + \mathcal{D}_{Kp,L}(Ku). \tag{7}$$

*The analogous identity holds for the loss of any fixed divergence. For independent product experiments with common parameter, Fisher tensors add. Relative-entropy losses add for product laws and product channels.*

*Proof.* Insert and subtract $F_{Kp}(Ku)$, or the intermediate divergence. For an independent product, the score is the sum of component scores. Each has mean zero, so cross terms vanish and the Fisher tensor adds. The logarithm of a product likelihood ratio is a sum; taking expectations proves relative-entropy additivity before and after product processing. □

*No-Go* 5.4 (Marginals and parameter-dependent processing). Neither arbitrary marginal addition nor parameter-dependent processing obeys the preceding rule.

*Proof.* For $x, y \in \{0, 1\}$, let $p_\theta(x, y) = \frac{1}{4}[1 + \theta(-1)^{x+y}]$, $|\theta| < 1$. Both marginals are uniform and have zero Fisher information, while the joint Fisher information is $1/(1-\theta^2)$. Dependence is observable in the joint law but not in separately supplied marginals. For the second claim take a constant input law and a channel whose every column is $(\theta, 1-\theta)^\top$. Its output is Bernoulli($\theta$) with positive Fisher information although the input has zero information. The channel has introduced parameter dependence rather than merely garbled a record. □

## 6 Finite loss and a canonical reference recovery

**Definition 6.1** (Reference-Bayes recovery). For a positive reference law $q$, define the stochastic map

$$R_{q,K}(x \mid y) = \frac{q_x K_{yx}}{(Kq)_y}. \tag{8}$$

It depends on $q$ and $K$, not on the unknown input $p$. It fixes the reference: $R_{q,K} K q = q$. It is canonical relative to this reference, not reference-free.

**Lemma 6.2** (Score operator and adjoint). *On mean-zero $L^2(q)$ define $Af = \mathbb{E}_q[f(X) \mid Y]$. Its adjoint is $(A^* b)_x = \sum_y K_{yx} b_y$. The operator $B = A^* A$ is self-adjoint and $0 \preceq B \preceq I$.*

*Proof.* Expand $\langle Af, b\rangle_{Kq}$ and interchange the finite sums to obtain the adjoint formula. Jensen gives $\|Af\|_{Kq} \le \|f\|_q$. Therefore $\langle f, Bf\rangle_q = \|Af\|_{Kq}^2$ lies between zero and $\|f\|_q^2$. □

**Theorem 6.3** (Finite chi-square loss controls recovery)**.** *For positive $p, q$, let*

$$\mathcal{L}_\chi(p, q; K) = \chi^2(p\|q) - \chi^2(Kp\|Kq).$$

*Then*

$$\mathcal{L}_\chi(p, q; K) = \mathcal{D}_{q,K}(p - q) \ge 0, \tag{9}$$

$$\sum_x \frac{[p_x - (R_{q,K}Kp)_x]^2}{q_x} \le \mathcal{L}_\chi(p, q; K), \tag{10}$$

$$\mathrm{TV}(p, R_{q,K}Kp) \le \tfrac{1}{2}\sqrt{\mathcal{L}_\chi(p, q; K)}. \tag{11}$$

*The factor $1/2$ and the square-root exponent in (11) are sharp in this class.*

*Proof.* Put $f = p/q - 1$, a mean-zero function. The two chi-square terms are $\|f\|_q^2$ and $\|Af\|_{Kq}^2$, so their difference is $\langle f, (I - B)f\rangle_q$ and equals (9). The recovered density relative to $q$ is $1 + Bf$, so the left side of (10) is $\|(I - B)f\|_q^2$. Since $\mathrm{spec}(B) \subset [0, 1]$, $(I - B)^2 \preceq I - B$. This proves (10). Cauchy–Schwarz gives $\sum_x q_x|(I - B)f_x| \le \|(I - B)f\|_q$, proving (11). For sharpness let $q = (1/2, 1/2)$, $p = (1/2 + t, 1/2 - t)$, and let $K$ erase the input. Then $R_{q,K}Kp = q$, $\mathcal{L}_\chi = 4t^2$, and $\mathrm{TV}(p, q) = |t|$. Equality holds in (11); no uniform exponent larger than $1/2$ can replace it near zero. □

**Theorem 6.4** (Relative-entropy loss, posterior loss, and recovery)**.** *Define $\mathcal{L}_D(p, q; K) = D(p\|q) - D(Kp\|Kq)$. With $r_p(x|y) = p_x K_{yx}/(Kp)_y$,*

$$\mathcal{L}_D(p, q; K) = \sum_y (Kp)_y D(r_p(\cdot|y)\|r_q(\cdot|y)), \tag{12}$$

$$D(p\|R_{q,K}Kp) \le \mathcal{L}_D(p, q; K), \tag{13}$$

$$\mathrm{TV}(p, R_{q,K}Kp) \le \sqrt{\mathcal{L}_D(p, q; K)/2}. \tag{14}$$

*The following are equivalent: zero relative-entropy loss; zero chi-square loss; equality of the two posteriors on every output; $R_{q,K}Kp = p$; existence of a stochastic recovery fixing both $p$ and $q$.*

*Proof.* Expand the posterior log ratio: $\log(r_p/r_q) = \log(p_x/q_x) - \log((Kp)_y/(Kq)_y)$ on every supported edge. Summing gives (12). Compare joint laws $P_{xy} = p_x K_{yx}$ and $Q_{xy} = (Kp)_y r_q(x|y)$. Their relative entropy is (12); their $X$ marginals are $p$ and $R_{q,K}Kp$. Log-sum processing proves (13); Lemma 2.2 gives (14). The posterior sum is zero iff each posterior agrees. This is equivalent to the edgewise likelihood-ratio condition $p_x/q_x = (Kp)_y/(Kq)_y$, also equivalent to zero chi-square loss by Theorem 5.1. Zero loss gives exact reference recovery by Theorem 6.3. Conversely reference recovery implies $(I - B)f = 0$ and hence zero chi-square loss. Finally any recovery $S$ fixing both laws gives $D(p\|q) \ge D(Kp\|Kq) \ge D(SKp\|SKq) = D(p\|q)$, so equality holds. The reference recovery itself supplies such an $S$ in the other direction. □

**Corollary 6.5** (All strictly convex finite losses have the same equality locus)**.** *For strictly convex $f$, equality in $D_f(Kp\|Kq) \le D_f(p\|q)$ is equivalent to the recovery conditions of Theorem 6.4.*

*Proof.* Equality in the Jensen step of Lemma 2.2 requires $p_x/q_x$ to be constant over every supported conditional output fibre. This is exactly the edgewise likelihood-ratio condition in the preceding proof. □

## 7 From infinitesimal defects to finite and global recovery

**Theorem 7.1** (Affine-mixture integral identity). *Let $p_t = q + t(p-q)$ and $v = p - q$. Then*

$$\mathcal{L}_D(p, q; K) = \int_0^1 (1-t)\mathcal{D}_{p_t,K}(v)\, dt. \tag{15}$$

*Consequently*

$$\mathrm{TV}(p, R_{q,K}Kp) \le \left[\tfrac{1}{2}\int_0^1 (1-t)\mathcal{D}_{p_t,K}(v)\, dt\right]^{1/2}. \tag{16}$$

*Proof.* Let $H(p) = \sum_x p_x \log p_x$. Its Hessian on the simplex is $F_p$. Taylor's integral formula gives $H(p) - H(q) - DH_q[v] = \int_0^1 (1-t)F_{p_t}(v)dt$. The left side is $D(p\|q)$. Apply the same formula to $Kp, Kq$ and subtract, using linearity of $K$. The recovery inequality is Theorem 6.4. □

**Corollary 7.2** (Relative Fisher loss along a mixture). *If $\mathcal{D}_{p_t,K}(v) \le \alpha^2 F_{p_t}(v)$ for all $t \in [0,1]$, then*

$$\mathrm{TV}(p, R_{q,K}Kp) \le \alpha\sqrt{D(p\|q)/2}.$$

*Proof.* Integrate the assumed inequality in (15); the integral of $F$ is $D(p\|q)$ by the preceding proof. Apply (14). □

*Remark* 7.3 (Why the mixture is additional structure). A parametric model need not contain $q + t(p-q)$. Theorem 7.1 is an ambient-law identity; using it from model-tangent measurements requires control on the entire mixture, not merely on the original model. The next theorem avoids that assumption by following actual model paths.

**Theorem 7.4** (Pathwise posterior-variation bound). *Let $p_t$, $0 \le t \le 1$, be a piecewise $C^1$ positive path on a fixed finite alphabet. Put $q_t = Kp_t$, $w = q_1$, and $d_t = \mathcal{D}_{p_t,K}(\dot{p}_t)$. Then the single base-reference recovery obeys*

$$\mathrm{TV}(p_1, R_{p_0,K}Kp_1) \le \tfrac{1}{2}\int_0^1 \sqrt{d_t C_t}\, dt, \qquad C_t = \sum_y \frac{w_y^2}{q_t(y)}. \tag{17}$$

*In particular it is at most $\frac{1}{2}\sqrt{\int_0^1 d_t dt}\sqrt{\int_0^1 C_t dt}$.*

*Proof.* Write $r_t(x|y) = p_t(x)K_{yx}/q_t(y)$. On supported edges, $\partial_t \log r_t(x|y) = s_t(x) - \bar{s}_t(y)$. Let $I_y(t) = \sum_x r_t(x|y)[\partial_t \log r_t(x|y)]^2$. Theorem 5.1 gives $d_t = \sum_y q_t(y) I_y(t)$. Since $\sum_x |\dot{r}_t(x|y)| \le \sqrt{I_y(t)}$, integration yields $\mathrm{TV}(r_1(\cdot|y), r_0(\cdot|y)) \le \frac{1}{2}\int_0^1 \sqrt{I_y(t)}dt$. Now $p_1 = \sum_y w_y r_1(\cdot|y)$, while $R_{p_0,K}Kp_1 = \sum_y w_y r_0(\cdot|y)$. Convexity of the $L^1$ norm and weighted Cauchy–Schwarz give $\sum_y w_y\sqrt{I_y(t)} \le \sqrt{\sum_y q_t(y)I_y(t)}\sqrt{\sum_y w_y^2/q_t(y)}$. This proves (17). A second Cauchy–Schwarz inequality proves the energy form. Positivity and compactness of each path ensure that its denominators stay positive; no uniform bound over an entire family is inferred without an extra hypothesis. □

**Theorem 7.5** (Connected-family sufficiency). *Let $\Theta$ be a piecewise-$C^1$ path-connected smooth parameter domain, and let $p_\theta > 0$ be $C^1$. Fix $\theta_0$. The following are equivalent:*

1. *$\mathcal{D}_{p_\theta,K}(Dp_\theta u) = 0$ for every $\theta$ and every tangent $u$;*
2. *$r_\theta(x|y)$ is independent of $\theta$ on every supported edge;*
3. *$R_{p_{\theta_0},K}Kp_\theta = p_\theta$ for all $\theta$;*
4. *one parameter-independent stochastic map recovers the whole family.*

*Proof.* Under (1), the derivative of every $\log r_\theta(x|y)$ vanishes along every model path by the equality condition in Theorem 5.1. Integrating along a path from $\theta_0$ to $\theta$ proves (2). Mixing the fixed posterior against $Kp_\theta$ gives (3), which implies (4). If (4) holds, Fisher contraction first by $K$ and then by the fixed recovery gives equality between the original and recovered Fisher tensors, proving (1). Equivalently, Theorem 7.4 with zero integrand proves (1)⇒(3). □

**Corollary 7.6** (Uniform path certificates give deficiency and risk bounds)**.** *Let $\mathcal{E} = (p_\theta)$ and $\mathcal{F} = (Kp_\theta)$. Suppose every $\theta$ has a declared path from $\theta_0$ for which the right side of (17) is at most $\eta$. With $\delta(\mathcal{F}, \mathcal{E}) = \inf_R \sup_\theta \mathrm{TV}(RKp_\theta, p_\theta)$,*

$$\delta(\mathcal{E}, \mathcal{F}) = 0, \qquad \delta(\mathcal{F}, \mathcal{E}) \le \min(1, \eta).$$

*For every loss in $[0,1]$, any input-experiment decision rule can be transferred through this recovery with risk increase at most $\eta$ uniformly in $\theta$.*

*Proof.* The known channel gives the first deficiency. The same base-reference map works on every path, so it witnesses the second bound. Composing an input decision rule with the recovery cannot increase total variation. Expectations of a $[0,1]$-valued loss differ by at most total variation, proving the risk statement. □

*No-Go* 7.7 (Pointwise equality and disconnected models). Fisher equality at one point is not global sufficiency. Connectedness in Theorem 7.5 cannot simply be dropped.

*Proof.* On a small interval containing 0 and 1/2, take

$$p_t = (1/4 + t/4 + t^2/8,\ 1/4 + t/4 - t^2/8,\ 1/2 - t/2)$$

and merge the first two outcomes. At $t = 0$ their scores agree, so the defect is zero. At $t = 1/2$ the defect is $25/429 > 0$. The within-block conditional law changes, so there is no common recovery. For disconnectedness take two components on each of which the law is constant, with respective values $(.2, .8)$ and $(.8, .2)$, and erase the record. Every tangent defect is zero, but one recovery from the same one-point output cannot produce both laws. □

## 8 Multiscale composition and acquisition

**Theorem 8.1** (Reference-consistent recovery composes exactly)**.** *For positive $q$ and composable $K, L$,*

$$R_{q,LK} = R_{q,K} R_{Kq,L}. \tag{18}$$

*For a chain $p_j = K_j p_{j-1}$, $q_j = K_j q_{j-1}$, let $\ell_j = D(p_{j-1}\|q_{j-1}) - D(p_j\|q_j)$. Then*

$$\mathrm{TV}\left(p_0, R_{q_0,K_1} \cdots R_{q_{k-1},K_k} p_k\right) \le \sqrt{\tfrac{1}{2} \sum_{j=1}^{k} \ell_j}. \tag{19}$$

*Proof.* The $(x, z)$ entry of the product in (18) is

$$\sum_y \frac{q_x K_{yx}}{(Kq)_y} \frac{(Kq)_y L_{zy}}{(LKq)_z} = \frac{q_x (LK)_{zx}}{(LKq)_z}.$$

Iterating proves that the composed recovery is the direct reference-Bayes recovery of the composite channel. The losses telescope to $D(p_0\|q_0) - D(p_k\|q_k)$. Apply Theorem 6.4 to the composite. □

**Corollary 8.2** (Arbitrary stage certificates)**.** *If arbitrary stochastic $S_j$ satisfy $\mathrm{TV}(p_{j-1}, S_j p_j) \le \eta_j$, then $\mathrm{TV}(p_0, S_1 \cdots S_k p_k) \le \sum_j \eta_j$.*

*Proof.* Insert $S_1 p_1$, use the triangle inequality and total-variation contraction by $S_1$, and repeat. The stronger square-root-of-total-loss bound requires the consistent references in Theorem 8.1; it is not inferred for arbitrary $S_j$. □

**Proposition 8.3** (Sequential acquisition uses conditional, not marginal, information)**.** *For a finite positive sequential law $p_\theta(z_{1:k}) = \prod_{j=1}^{k} p_\theta(z_j \mid z_{<j})$, the full-record Fisher tensor is*

$$F_{1:k} = \sum_{j=1}^{k} \mathbb{E}_\theta[F_{Z_j|Z_{<j}}].$$

*Relative entropy has the analogous conditional chain rule. These formulas allow adaptive designs whose policies are parameter independent conditional on the recorded history.*

*Proof.* The full score is $\sum_j s_j$. Differentiating each conditional normalization gives $\mathbb{E}[s_j \mid Z_{<j}] = 0$. For $i < j$, $s_i$ is measurable with respect to $Z_{<j}$, so $\mathbb{E}[s_i s_j^\top] = 0$. Expanding the score covariance proves the formula. Expanding the log likelihood ratio into conditional log ratios proves the relative-entropy chain rule. A recorded adaptive policy contributes no score of its own if its conditional action probabilities do not depend on the unknown parameter. □

## 9 Global distances do not reconstruct the experiment

**Theorem 9.1** (Ambient Fisher–Rao distance and completion)**.** *On the full open simplex,*

$$d_F(p, q) = 2 \arccos \sum_x \sqrt{p_x q_x}. \tag{20}$$

*Its metric completion is the closed simplex in square-root coordinates. Every stochastic channel contracts this distance.*

*Proof.* The map $p \mapsto 2(\sqrt{p_1}, \ldots, \sqrt{p_n})$ lies on the radius-two sphere and has squared differential norm $\sum_x u_x^2/p_x = F_p(u)$. The shorter great-circle arc between positive vectors remains positive: its coefficients in the two endpoint vectors are positive. Thus its length is the spherical distance (20). Closure gives the nonnegative spherical orthant, identified with the closed simplex. For each $y$, Cauchy–Schwarz gives $\sum_x K_{yx}\sqrt{p_x q_x} \le \sqrt{(Kp)_y (Kq)_y}$. Summing shows that affinity increases; arccos decreases, proving contraction. □

*Remark* 9.2 (Ambient versus model-intrinsic distance). For a restricted law image, the intrinsic distance uses only paths inside that image and can exceed (20). It may even be infinite between disconnected components. An ambient great-circle path is not automatically an admissible model path.

*No-Go* 9.3 (The entire Fisher field is not a complete experiment invariant). Two positive finite experiments on the same parameter interval can have identical Fisher tensors at every parameter and fail to be Blackwell equivalent.

*Proof.* For $0 < \theta < \pi$, let $\mathcal{E}_\theta$ be Bernoulli with success probability $\sin^2(\theta/2)$. Its Fisher information is 1. Let $\mathcal{G}_\theta$ consist of two independent Bernoulli variables with success probability $\sin^2(\theta/(2\sqrt{2}))$. Each contributes $1/2$, so its total Fisher information is also 1. For parameters separated by $d$, their Hellinger affinities are respectively $\cos(d/2)$ and $\cos^2(d/(2\sqrt{2}))$. At $d = 1$ these are approximately 0.87758256 and 0.88012230, which differ. Mutual randomization would force equality of affinity by processing in both directions, contradicting this difference. Thus even the complete parameterized Fisher field does not determine the global statistical experiment. □

*No-Go* 9.4 (Fisher does not select a unique finite divergence). Markov contraction and the same local Fisher Hessian do not select one finite divergence.

*Proof.* The convex functions

$$f_1(t) = t\log t - t + 1, \quad f_2(t) = \tfrac{1}{2}(t-1)^2, \quad f_3(t) = 2(\sqrt{t}-1)^2$$

all have $f(1) = f'(1) = 0$ and $f''(1) = 1$. Taylor expansion gives $D_f(p + tu\|p) = \frac{1}{2}t^2 F_p(u) + o(t^2)$ for all three. They contract by Lemma 2.2. At $p = (.8, .2)$ and $q = (.3, .7)$ their values are approximately 0.53411081, 0.59523810, and 0.54374525, respectively. They therefore agree infinitesimally and disagree finitely. □

## 10 Singularities, support, and derivative control

*No-Go* 10.1 (A Fisher-null tangent need not be an exact gauge direction). At singular points, $\ker Dp$ need not describe a locally constant law direction; even all finite jets can miss a distinction.

*Proof.* For $p_\theta = (1/2 + \theta^2, 1/2 - \theta^2)$ near zero,

$$F_\theta = \frac{16\theta^2}{1 - 4\theta^4}, \qquad F_0 = 0, \qquad \mathrm{TV}(p_\theta, p_0) = \theta^2 > 0 \quad (\theta \neq 0).$$

The second jet sees a distinction invisible to the first. More strongly put $a(\theta) = e^{-1/\theta^2}$ for $\theta \neq 0$ and $a(0) = 0$, and use $p_\theta = (1/2 + a(\theta), 1/2 - a(\theta))$ on a sufficiently small interval. Every derivative of $a$ at zero vanishes, yet $p_\theta \neq p_0$ off zero. No finite-jet or even formal Taylor rule detects every smooth distinction without an analyticity or other rigidity assumption. □

**Example 10.2** (Divergent coordinates, finite boundary distance)**.** For Bernoulli($\theta$), $F_\theta = 1/[\theta(1-\theta)]$ diverges at zero, but its Fisher distance to the boundary is $2\arcsin\sqrt{\theta} \to 0$. A divergent coordinate coefficient is not infinite intrinsic distance. The square-root completion of Theorem 9.1 does not assert a smooth positive-definite tensor across all support strata.

*No-Go* 10.3 (Uniform law closeness and small deficiency do not control Fisher). Uniform total-variation or Hellinger convergence of experiments, even with vanishing Le Cam distance, does not by itself imply convergence of Fisher tensors.

*Proof.* On a fixed interval let $p_{n,\theta}$ be Bernoulli with probability $1/2 + n^{-1}\sin(n^2\theta)$, $n \geq 4$, and let $p_{0,\theta}$ be constant Bernoulli(1/2). Uniformly in $\theta$, $\mathrm{TV}(p_{n,\theta}, p_{0,\theta}) \leq 1/n$, so identity randomizations give both deficiencies at most $1/n$. Hellinger distance also tends uniformly to zero since squared Hellinger distance is at most twice total variation. But $F_n(0) = 4n^2$ while $F_0(0) = 0$. Derivative control is missing. This does not contradict a theorem that *separately assumes* convergence or uniform control of square-root derivatives. □

**Theorem 10.4** (Square-root derivative stability)**.** *Let $s(\theta) = \sqrt{p_\theta}$ and $\widetilde{s}(\theta) = \sqrt{\widetilde{p}_\theta}$ be differentiable on the same chart. At a point write $J = Ds$, $\widetilde{J} = D\widetilde{s}$. If $\|J\| \leq M$ and $\|\widetilde{J} - J\| \leq \eta$, then*

$$\|\widetilde{F} - F\| \leq 4\eta(2M + \eta). \tag{21}$$

*All norms are operator norms in a fixed declared parameter comparison metric.*

*Proof.* The Fisher matrices are $4J^\top J$ and $4\widetilde{J}^\top\widetilde{J}$. Put $E = \widetilde{J} - J$ and expand the difference as $4(J^\top E + E^\top J + E^\top E)$. Submultiplicativity gives (21). □

**Proposition 10.5** (Nuisance minimization preserves contraction)**.** *Fix a common target/nuisance splitting. For positive-semidefinite forms $F^{\mathrm{out}} \preceq F^{\mathrm{in}}$, define $F^{\mathrm{eff}}(u) = \inf_v F((u,v))$. Then $F^{\mathrm{out,eff}} \preceq F^{\mathrm{in,eff}}$. When the nuisance block is positive definite this is the usual Schur complement.*

*Proof.* For every nuisance vector $v$, $F^{\mathrm{out}}((u,v)) \leq F^{\mathrm{in}}((u,v))$. Evaluate the left infimum at an arbitrarily near-minimizer of the right infimum and let its error tend to zero. Completing the square in the nuisance variables gives the Schur complement when that block is positive definite. Continuity under rank-changing nuisance conventions is not part of this argument. □

## 11 Resolution gates: what can and cannot be maximal

**Theorem 11.1** (A spectral margin gives projector stability)**.** *Let $A, B$ be symmetric matrices in one fixed comparison inner product. Suppose* $\operatorname{dist}(\tau, \operatorname{spec} A) \geq \gamma > 0$ *and* $\|A - B\| \leq \eta < \gamma$*. Let* $P = \mathbf{1}_{(\tau,\infty)}(A)$ *have rank* $r$*,* $0 < r < d$*, and let* $Q = \mathbf{1}_{(\tau,\infty)}(B)$*. Then* $\operatorname{rank} Q = r$ *and*

$$\|P - Q\|_{\mathrm{Frob}} \leq \frac{\sqrt{2r}\,\eta}{2\gamma - \eta}. \tag{22}$$

*For a fixed positive comparison matrix $G$, apply the statement to $G^{-1/2} F G^{-1/2}$. A changing $G$ needs its own perturbation bound.*

*Proof.* The variational min–max characterization implies that each ordered eigenvalue moves by at most $\eta$: every unit Rayleigh quotient changes by at most $\eta$. Thus no eigenvalue crosses $\tau$ and the ranks agree. Let $u_j$ be an orthonormal eigenbasis of the high $B$ sector, with eigenvalues $b_j \geq \tau + \gamma - \eta$. In the low $A$ sector all eigenvalues satisfy $a_i \leq \tau - \gamma$. Projecting $(A - b_j I)u_j = (A - B)u_j$ onto that sector and expanding in its eigenbasis gives $\|(I - P)u_j\| \leq \eta/(2\gamma - \eta)$. Sum squares over $j = 1, \ldots, r$. For equal-rank orthogonal projectors, $\|P - Q\|_{\mathrm{Frob}}^2 = 2\|(I - P)Q\|_{\mathrm{Frob}}^2$, which proves (22). The cases $r = 0, d$ have $P = Q$ under the same gap assumption. □

*No-Go* 11.2 (Approximate Rayleigh gates need not have a greatest subspace). For a positive loss operator $D$, the condition $\langle u, Du\rangle \leq a\|u\|^2$ on every $u$ in a subspace need not admit a greatest admissible subspace.

*Proof.* Take $D = \operatorname{diag}(0, 2)$ and $a = 1$. Each line spanned by $(1, 1)$ or $(1, -1)$ is admissible. A greatest admissible subspace would contain both and hence all of $\mathbb{R}^2$. But $(0, 1)$ has Rayleigh quotient 2, a contradiction. Replacing $D$ by $D/2$ and $a$ by $1/2$ puts the example inside the $[0, 1]$ spectrum of a normalized Fisher-loss operator. For an explicit tangent realization, take two independent uniform bits, retain only the first, and restrict the mean-zero score space to the span of the two individual sign scores. The normalized loss operator on that span is $\operatorname{diag}(0, 1)$. □

**Theorem 11.3** (Correct maximality contract)**.** *For self-adjoint $D$, the spectral sector $W_a = \operatorname{im} \mathbf{1}_{(-\infty,a]}(D)$ is the unique greatest $D$-*invariant *subspace satisfying the Rayleigh bound. At $a = 0$ and $D \succeq 0$, the exact zero-loss sector is* $\ker D$ *and is greatest even without invariance.*

*Proof.* $W_a$ is invariant and satisfies the bound. If an invariant subspace $U$ contained a component in an eigenspace with eigenvalue above $a$, a polynomial in $D$ isolating that eigenvalue would produce a nonzero vector of $U$ entirely in that eigenspace, violating the bound. Thus $U \subset W_a$. For $a = 0$, $\langle u, Du\rangle = 0$ implies $D^{1/2}u = 0$, hence $u \in \ker D$. □

*Remark* 11.4 (Multiple gates and transport). Intersections of individually declared spectral sectors are sound subspaces, but they need not be complete among all subspaces satisfying several unrestricted Rayleigh bounds. Noncommuting loss operators require an explicit common-invariance or other admissibility contract. Relabeling outcomes transports all formulas naturally, and reference-consistent garblings have the recovery composition in Theorem 8.1. None of these statements selects a latent connection or an undeclared holonomy class.

**Proposition 11.5** (Finite-grid certification with a declared modulus)**.** *Let $d(\theta)$ be a scalar gate diagnostic on a compact metric domain, known to be $L$-Lipschitz. If an $h$-net has certified upper bounds $d(\theta_j) \leq b$, then $d(\theta) \leq b + Lh$ everywhere. If the grid bounds fail with probabilities at most $\alpha_j$, the uniform statement fails with probability at most $\sum_j \alpha_j$.*

*Proof.* Choose a grid point within $h$ and apply the Lipschitz inequality. The probability statement is the union bound. The modulus is an assumption to certify independently; the grid alone cannot identify it, by No-Go 4.9. □

## 12 A dominated finite-partition extension

**Theorem 12.1** (Partition exhaustion of Fisher energy). *Let $(\Omega, \mathcal{A}, P)$ admit increasing finite partitions generating $\mathcal{A}$ modulo $P$-null sets. Let $s \in L^1(P)$ have mean zero, representing a dominated signed tangent $d\dot{P} = s\,dP$. For partition $\mathcal{P}_j$ set $s_j = \mathbb{E}[s \mid \mathcal{P}_j]$ and $F_j = \int s_j^2 dP$. Then*

$$F_j \uparrow \int s^2 dP \in [0, \infty].$$

*If $s \in L^2(P)$ and $K$ is a parameter-independent channel, the score-variance identity remains valid with conditional expectations in $L^2$.*

*Proof.* Conditional Jensen gives $F_j \le F_{j+1}$ and $F_j \le \int s^2 dP$ when the latter is finite. The upward martingale convergence theorem gives $s_j \to s$ almost surely and in $L^1$; for $s \in L^2$, convergence is also in $L^2$. Thus $F_j \to \int s^2 dP$ in the finite case. In the infinite case Fatou gives $\infty = \int s^2 dP \le \liminf_j F_j$, so the same conclusion holds. The martingale convergence theorem is an explicitly imported standard measure-theoretic input. In $L^2$, conditional expectation is an orthogonal projection, and the Pythagorean identity gives the variance formula exactly as in Theorem 5.1. □

**Corollary 12.2** (Partition-continuous selection). *Suppose a metric on these dominated tangent directions is required to be the limit of its finite-partition values, with the same exact finite Čencov normalization $c$. It is necessarily $c \int s^2 dP$. Under the uniform approximate assumptions of Theorem 4.3, the lower and upper limits of partition metric values lie between $e^{-2\varepsilon} c \int s^2 dP$ and $e^{2\varepsilon} c \int s^2 dP$.*

*Proof.* Apply the exact or quantitative finite theorem to the partition law and its tangent, discard zero-probability cells, and pass to the limit using Theorem 12.1. For one-cell partitions both tangent energies are zero. The approximate theorem bounds lower and upper limits; it does not by itself assert existence of a metric limit. □

*Remark* 12.3 (What this does not extend). This is a dominated-tangent, partition-continuity result, not an unrestricted characterization on every infinite-dimensional or non-dominated experiment. For example the family $\delta_\theta$ has perfectly separated distinct laws but no nontrivial dominated location score of this kind. Its equality-of-law quotient and testing comparisons remain meaningful; the displayed Fisher module does not apply.

## 13 Quantum boundary: a valid bridge and an invalid uniqueness claim

**Imported Theorem 13.1** (Petz monotone-metric classification). *On faithful finite-dimensional density matrices, CPTP-monotone Riemannian metrics form a family indexed by symmetric normalized operator-monotone functions. They agree with classical Fisher on commuting tangent data but need not agree on noncommuting tangents [7]. This classification is imported, not reproved here.*

**Proposition 13.2** (Quantum descent keeps every Petz metric and selects none). *Quantum lossless re-descriptions are CPTP maps with a CPTP left inverse. Examples are isometric embeddings $\rho \mapsto V\rho V^\dagger$ and appending a fixed ancilla, $\rho \mapsto \rho \otimes \tau$. Every CPTP-monotone metric is invariant under them, by the argument of Step 6 in Theorem 3.4 with Markov replaced by CPTP.* Executed (G4):

- *SLD, BKM, WY and RLD are each invariant under $\rho \mapsto V(\rho \otimes \tau)V^\dagger$ (100/100 cases);*
- *they differ by a relative spread of up to* 16.6 *on noncommuting tangents;*

- *on commuting tangents they coincide exactly with Fisher, so the classical theorem is recovered.*

*Descent is therefore not enough in quantum theory, and the task must declare the metric. This is RG's task lock (L39).*

**Proposition 13.3** (A concrete noncommuting separation). *At* $\rho = \mathrm{diag}(.8, .2)$ *and* $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, *in the normalization agreeing with classical Fisher,*

$$g_{\mathrm{SLD},\rho}(A, A) = 4, \qquad g_{\mathrm{BKM},\rho}(A, A) = \frac{2 \log 4}{.6} \simeq 4.62098120.$$

*Proof.* The SLD coefficient in the eigenbasis is $2/(\lambda_i + \lambda_j)$ and the BKM coefficient is $(\log \lambda_i - \log \lambda_j)/(\lambda_i - \lambda_j)$. There are two unit off-diagonal entries of $A$. Substitution gives the two values. For diagonal tangents both coefficients reduce to $1/\lambda_i$, explaining their classical agreement. □

**Theorem 13.4** (Quantum entropy-Hessian mixture identity). *For faithful states* $\rho, \sigma$, *use Umegaki relative entropy* $D(\rho\|\sigma) = \mathrm{tr}[\rho(\log \rho - \log \sigma)]$. *Put* $A = \rho - \sigma$ *and* $\rho_t = \sigma + tA$. *Define*

$$g_{\mathrm{BKM},\rho}(A, A) = \int_0^\infty \mathrm{tr}[A(\rho + sI)^{-1} A (\rho + sI)^{-1}]\, ds.$$

*Then*

$$D(\rho\|\sigma) = \int_0^1 (1 - t) g_{\mathrm{BKM},\rho_t}(A, A) dt.$$

*For a fixed CPTP map* $\Phi$ *whose output states are faithful, subtracting the output identity expresses relative-entropy loss as the integral of the BKM contraction defect.*

*Proof.* The matrix-log resolvent formula gives $D \log_\rho[A] = \int_0^\infty (\rho + sI)^{-1} A (\rho + sI)^{-1} ds$. It follows by differentiating the resolvent representation of $\log \rho$; faithfulness justifies differentiation and integration in finite dimension. For $H(\rho) = \mathrm{tr}(\rho \log \rho)$, spectral differentiation under the trace gives $DH_\rho[A] = \mathrm{tr}[A(\log \rho + I)]$, and therefore $D^2 H_\rho[A, A] = \mathrm{tr}[AD \log_\rho[A]] = g_{\mathrm{BKM},\rho}(A, A)$. Taylor's integral formula gives the stated identity because the Bregman remainder of $H$ is $D(\rho\|\sigma)$ and $\mathrm{tr}\, A = 0$. Linearity of $\Phi$ sends the affine mixture to its output affine mixture, so subtraction proves the final assertion. Nonnegativity of the BKM defect follows from its imported monotone-metric status, not from a new classification proof. □

*Remark* 13.5 (No classical recovery formula is transplanted). The quantum identity does not prove that an unrotated Petz recovery has the classical bound (13). The classical proof used conditional probabilities and log-sum on a joint law, which are not available in that form for noncommuting states. A quantum recovery theorem, its recovery-map choice, and its norm or fidelity bound must be imported or separately proved. The present quantum result is an entropy-Hessian bridge, not closure of all quantum recovery gaps.

## 14 Why Resolution Geometry exists because of Čencov

RG has two halves. The first is what descent forces. The second is what it must add where Čencov provably stops.

| RG component | Relation to Čencov | Where |
|---|---|---|
| Quotient by observational equivalence; descent rule | *is* Čencov's invariance hypothesis | Lemma 3.2, Theorem 3.4 |

| RG component | Relation to Čencov | Where |
|---|---|---|
| Classical metric layer (Fisher, unit anchor) | *forced* | Corollary 3.6 |
| Information loss and recovery calculus | built on the Fisher defect | Sections 5–8 |
| Global experiment comparison (Blackwell, deficiency) | needed: Fisher fields do not determine experiments | No-Go 9.3 |
| Finite divergences and losses | needed: the local metric selects no finite divergence | No-Go 9.4 |
| Walls, strata, rooms | needed: null tangents are not exact fibres | No-Go 10.1 |
| Quantum task lock (L39) | needed: descent keeps every Petz metric | Proposition 13.2 |
| Finite-data certification | needed: no finite panel certifies invariance | No-Go 4.9 |

The thesis is that RG exists primarily because of Čencov. Descent through the observational quotient is the one rule from which the classical geometry follows uniquely. Every further structure of RG answers a precise point where that uniqueness provably ends.

## 15 Assembly theorem and coverage ledger

**Theorem 15.1** (RG quantitative completion in the declared classical scope)**.** *Fix a positive finite-alphabet experiment, parameter-independent channels, a reference law, and a complete RG comparison and gate contract. Assume the universal metric hypotheses of Assumption 4.1 when invoking metric selection. Then:*

1. *the selected metric is quantitatively Fisher as in (4), or as in Theorem 4.4 when symmetry is approximate;*
2. *exact and finite processing losses have the variance, recovery, and mixture identities of Sections 5–7;*
3. *zero model-tangent loss implies one global recovery on a path-connected positive family, and uniform path bounds imply the deficiency certificate of Corollary 7.6;*
4. *reference-consistent multiscale recoveries compose and losses telescope;*
5. *derivative control and a fixed positive spectral margin imply the stated metric and projector stability bounds;*
6. *exact lossless sectors are greatest kernels; approximate spectral sectors are greatest only in their stated invariant-subspace contract.*

*No item determines a unique finite divergence, a global likelihood embedding, a singular smooth tensor, a latent connection, a quantum metric, or an observation-independent ontology.*

*Proof.* Apply Theorems 4.3, 4.4 and Corollary 4.6 for (1). Theorem 5.1, Lemma 6.2, Theorems 6.3, 6.4, and 7.1 prove (2). Theorems 7.4 and 7.5 and Corollary 7.6 prove (3). Proposition 5.3 and Theorem 8.1 prove (4). Theorems 10.4 and 11.1 prove (5). Corollary 5.2, No-Go 11.2, and Theorem 11.3 prove (6). The non-claims follow from the displayed counterexamples and from the absence of any premise selecting the remaining structures. □

Table 2: Coverage is explicit and scoped, not an assertion that all RG mathematics is closed.

| Gap or requested implication | Result here | Boundary or remaining input |
|---|---|---|
| Exact metric uniqueness | Corollary 4.5, rederived | All finite dimensions and sufficient embeddings |
| Approximate metric uniqueness | Theorem 4.3, $e^{\pm 2\varepsilon}$ | Theorem 4.4 relaxes symmetry; universal uniform premise |
| Meaning of information loss | Theorem 5.1 | Parameter-independent processing |
| Finite recovery from local data | Theorems 6.3, 7.1, 7.4 | Full mixture or path control, not one tangent snapshot |
| Global sufficiency | Theorem 7.5 | Connected positive model; all tangent directions |
| Approximate experiment equivalence | Corollary 7.6 | Uniform path and output-overlap bounds |
| Multiscale closure | Theorem 8.1 | Reference consistency; no arbitrary decoder substitution |
| Global metric versus experiment | No-Go 9.3 | Likelihood embedding remains necessary |
| Finite divergence uniqueness | No-Go 9.4 | An additional operational selection principle |
| Singular null versus exact fibre | No-Go 10.1 | Higher-order or full-law data; smooth jets can all fail |
| Boundary continuation | Theorem 9.1 | Metric completion, not a universal smooth tensor |
| Deficiency-to-Fisher continuity | No-Go 10.3; Theorem 10.4 | Derivative control must be added |
| Nuisance and spectral stability | Proposition 10.5; Theorem 11.1 | Common splitting, comparison metric, and margins |
| Greatest approximate sector | No-Go 11.2; Theorem 11.3 | Invariance or another greatest-subobject hypothesis |
| Infinite observation alphabets | Theorem 12.1 | Dominated tangent and partition-continuity contract |
| Quantum metric selection | Imported Theorem 13.1; Proposition 13.3 | No unique quantum metric follows |
| Quantum finite-scale bridge | Theorem 13.4 | Recovery theorem remains separate |
| Finite-data universal certification | No-Go 4.9; Proposition 11.5 | Declared regularity and confidence bounds |

## 16 Computational stress battery and its limits

The companion scripts generate synthetic mathematical examples, not laboratory data. The source-defined design is not an externally preregistered protocol. No numerical test is used to prove a universal theorem or to certify Assumption 4.1.

| Family | Cases |
|---|---|
| Random positive channels, laws, and score directions | 800 |
| Affine-mixture Fisher-loss integrals | 100 |
| Smooth positive path recovery bounds | 100 |
| Rational splitting and approximate metric examples | 200 |
| Non-permutation-invariant approximate metric examples | 200 |
| Exactly sufficient connected family parameter values | 91 |
| Square-root exponent witness values | 21 |
| Oscillatory deficiency/Fisher counterexample values | 6 |
| Spectral projector perturbations | 200 |
| Efficient-information contractions | 100 |
| Square-root derivative perturbations | 100 |
| Classical numerical CSV rows | 1918 |
| Exact rational partition/law cases | 222 |
| Quantum BKM mixture/depolarization cases | 60 |

The exact partition cases enumerate all set partitions on $n = 2, 3, 4, 5$ outcomes, with counts $2, 5, 15, 52$, and three rational input laws per partition. They verify variance and deterministic-recovery identities using rational arithmetic. Additional symbolic checks verify normalized divergence Hessians, the singular Fisher formula, the entropy Hessian, rational-splitting coefficients, and reference-Bayes composition. They are not a formal proof assistant development.

| Executed aggregate check | Result |
|---|---|
| Classical numerical assertions | 28/28 |
| Exact rational/symbolic assertion groups | 8/8 |
| Quantum matrix assertion groups | 3/3 |
| Counterexample/mutation canaries within classical checks | 15/15 detected |
| Glue G1: Markov left inverse $\iff$ disjoint supports (LP) | 400/400 |
| Glue G2: necessity canaries (permutation-only, dimension scale) | detected |
| Glue G3: Fisher splitting-invariant / channel-monotone | 500/500, 500/500 |
| Glue G4: quantum descent invariance / spread / commuting collapse | 100/100, 16.6, 0 |

The canaries reject: declaring a destructive merge lossless; substituting an unrelated decoder; promoting pointwise equality to global sufficiency; reconstructing an experiment from Fisher alone; inferring Fisher stability from deficiency; unrestricted greatest-subspace claims; adding marginal information without a joint law; selecting a unique finite divergence from its Hessian; linear recovery scaling in quadratic loss; parameter-dependent garbling; projector stability without a margin; equating a null tangent with exact gauge; dropping connectedness; equating a divergent coordinate metric with infinite distance; and transplanting classical metric uniqueness to noncommuting quantum data.

| Computed diagnostic | Value |
|---|---|
| Maximum normalized conditional-variance identity residual | $5.085 \times 10^{-16}$ |
| Maximum affine-mixture integral absolute residual | $4.441 \times 10^{-16}$ |
| Maximum reference-recovery composition entry residual | $3.331 \times 10^{-16}$ |
| Maximum independent-product KL-loss absolute residual | $6.662 \times 10^{-16}$ |
| Largest path actual-TV / proved-bound ratio | 0.9198932 |
| Largest projector-error / proved-bound ratio | 0.351522 |
| Fitted exponent in the square-root witness | 0.5000000000000004 |
| Maximum quantum BKM integral absolute residual | $3.109 \times 10^{-15}$ |

Residual figures above are rounded upward for display where appropriate; the JSON files retain the raw floating-point values. Negative maximum inequality violations in the JSON indicate positive

slack, not an error. The fitted exponent illustrates an analytic witness whose exponent is already proved in Theorem 6.3.

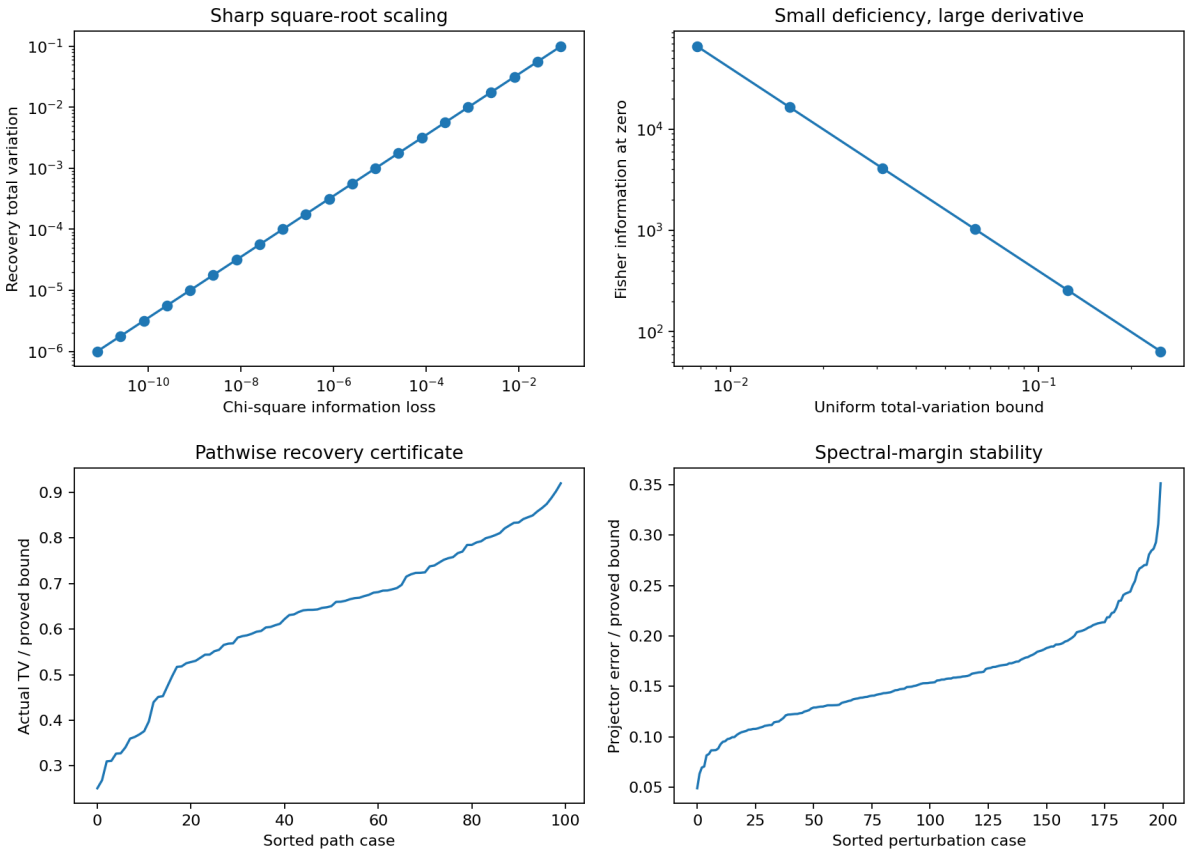



Figure 1: Computed synthetic diagnostics: square-root recovery scaling; small law distance with growing Fisher derivative; pathwise recovery certificates; and projector stability with a positive margin. These are quantitative plots of executed examples, not conceptual illustrations or evidence about physical systems.

**Secondary consistency check.** The release package includes a verifier that checks source hashes, aggregate assertion counts, CSV row counts, independent recomputation of selected maxima, and a fresh rerun comparison in a separate output directory. The executed verifier passed 27/27 checks, including fresh-process agreement of all three result JSON files and both numerical CSV files. This checks artifact consistency, not independent mathematical correctness. Universal proof review remains necessary.

## 17 Open problems and publication positioning

**Open Problem 17.1** (Sharp quantitative characterization)**.** Determine the optimal distortion constants in Theorems 4.3 and 4.4. The bounds 2 and 4 are not claimed sharp. Develop dimension-limited or restricted-splitting variants with explicit regularity assumptions.

**Open Problem 17.2** (Finite-panel to universal invariance)**.** Find useful regularity and dimension-growth assumptions under which a finite invariance panel yields a certified global distortion bound. No-Go 4.9 excludes a premise-free solution.

**Open Problem 17.3** (Sharp model-path recovery)**.** Optimize the path certificate over model paths and reference laws; identify necessary overlap conditions for uniform bounds near support

walls. Theorem 7.4 is sufficient and explicit, not an optimal characterization of deficiency.

**Open Problem 17.4** (Singular and quantum extensions)**.** Develop appropriate support-stratified and non-dominated comparison modules, and state quantum recovery results with their correct map and norm choices. Neither classical sufficient embeddings nor commuting agreement selects a universal quantum metric.

**Open Problem 17.5** (Is continuity necessary?)**.** Does there exist a family satisfying descent (R1) and positivity (R3), but not continuity, that is not $cF$? The splitting and merging relations connect every law to a uniform law, but they constrain metric values only on finitely many tangent lines per intermediate law. A positive answer would show that RG's continuity declaration is essential. A negative answer would remove it.

**Open Problem 17.6** (Independent proof and priority review)**.** Compare each theorem, particularly Theorems 4.3 and 7.4, against the quantitative Čencov, statistical sufficiency, strong data-processing, and approximate experiment-comparison literature. Establish which statements are new, which are known corollaries, and whether sharper hypotheses or constants already exist. This manuscript does not certify that review.

**Conclusion.** Čencov's theorem follows in full from RG's descent rule, because lossless redescriptions are exactly congruent splittings. RG's classical geometry is therefore forced, and its further machinery is required exactly where uniqueness ends. Beyond that, the strongest defensible improvement is not "Fisher has been superseded." It is that a classical uniqueness theorem can be placed inside a quantitative, finite-scale, reference-consistent recovery calculus. The positive results close the displayed finite classical implications under explicit hypotheses. The no-go results close several tempting but false implications by showing exactly why they cannot be proved. The remaining infinite-dimensional, quantum, finite-data, and priority questions remain visible rather than being relabeled as locks.

# A Dependency chain for proof review

The following is the logical dependency map; no computational assertion is a premise.

1. Convexity of $MK = I \Rightarrow$ Lemma 3.2; with Lemma 4.2, uniform splittings, density, continuity and Theorem 5.1 $\Rightarrow$ Theorem 3.4 $\Rightarrow$ Corollary 3.6; explicit families $\Rightarrow$ Proposition 3.7.
2. Convexity, finite sums, and Cauchy–Schwarz $\Rightarrow$ Lemma 2.2.
3. Permutation symmetry $\Rightarrow$ Lemma 4.2; two common-refinement comparisons and continuity $\Rightarrow$ Theorem 4.3 $\Rightarrow$ Corollaries 4.5, 4.6 and Proposition 4.7.
4. Conditional score formula $\Rightarrow$ Theorem 5.1 $\Rightarrow$ Corollary 5.2 and Proposition 5.3.
5. Conditional-expectation adjoint $\Rightarrow$ Lemma 6.2 $\Rightarrow$ Theorem 6.3.
6. Posterior log-ratio expansion and log-sum $\Rightarrow$ Theorem 6.4 $\Rightarrow$ Corollary 6.5.
7. Entropy Hessian and Taylor's integral formula $\Rightarrow$ Theorem 7.1 $\Rightarrow$ Corollary 7.2.
8. Posterior score derivative and two Cauchy–Schwarz inequalities $\Rightarrow$ Theorem 7.4 $\Rightarrow$ Theorem 7.5 and Corollary 7.6.
9. Bayes matrix multiplication and telescoping $\Rightarrow$ Theorem 8.1; total-variation contraction $\Rightarrow$ Corollary 8.2.
10. Conditional score martingale differences $\Rightarrow$ Proposition 8.3.

11. Square-root spherical embedding ⇒ Theorem 9.1; explicit examples ⇒ No-Gos 9.3, 9.4, 10.1, 10.3.

12. Matrix expansion, variational minimization, and spectral min–max ⇒ Theorem 10.4, Proposition 10.5, and Theorem 11.1.

13. Two-line obstruction and invariant spectral decomposition ⇒ No-Go 11.2 and Theorem 11.3.

14. Imported martingale convergence plus conditional Jensen ⇒ Theorem 12.1 ⇒ Corollary 12.2.

15. Imported Petz classification ⇒ the quantum nonuniqueness context; matrix-log resolvent differentiation ⇒ Theorem 13.4.

16. The preceding scoped results ⇒ Theorem 15.1. No theorem selects undeclared RG protocol data.

# B Reproducibility and Python appendix

The package contains the complete executable sources rather than pseudocode:

- `rg_cencov_battery.py`: classical numerical families and mutation canaries;
- `rg_cencov_exact.py`: exact rational partitions and symbolic identities;
- `rg_cencov_quantum.py`: faithful-state BKM mixture checks;
- `rg_cencov_verify.py`: artifact checks and independent rerun comparison;
- `rg_cencov_spec.json`, result JSON files, classical and quantum CSV files, and the computed figure;
- this LaTeX source, the compiled PDF, a README, environment requirements, and a SHA-256 manifest.

The classical random-channel seeds are $101, 202, 303, 404, 505$; supplementary classical cases use seed 606; quantum cases use 707. Numerical thresholds and exact-case counts are encoded in the sources and result files. The robust-metric examples test an explicitly admissible family, not the space of all possible metric families. Exact partition enumeration covers deterministic channels only; the random numerical suite covers non-deterministic channels as well. Smooth flat-germ and finite-panel bump counterexamples are analytic constructions, not claims of numerical detection at arbitrary resolution.

**Build.** Run the three Python scripts and the verifier in the package directory, then compile `rg_cencov_stability.tex` twice with pdfLaTeX. The PDF uses only the included quantitative figure and standard LaTeX packages. The README provides the exact commands and distinguishes numerical reproduction from proof verification.

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